HCov Exam Sheet 2025/26 - The Gratwick Files

Theorem: Random Theorems

- **Triangle Inequality** $|z+w| \le |z|+|w|$
- **Reverse Triangle Inequality** $|z-w| \ge ||z|-|w||$
- **Convergent Geometric Series** For $|r|<1$, $\frac{1}{1-r} = \sum_{k=0}^{\infty} r^k$

Remark: Notation

Look at the context and these are usually obvious

- $z_0 = x_0 + iy_0, w = a+ib$
- $S, U, D \subseteq \mathbb{C}$
- $f(z) = u(x,y) + iv(x,y)$
- $f: S \to C$, $D \to C$, etc (context :)))
- z, z_0, w etc $\in \mathbb{C}, S, D$ etc
- Γ is a contour/curve (context :)))
- $[a,b] \subseteq \mathbb{R}$
- When mapping A 'onto' B, all of B must be in the image of the map

Complex Fundamentals

Definition: Basics

- $\mathbb{C} = \{x+iy \mid x,y \in \mathbb{R}\}$ is a field
- $\Re(z) = \frac{z+\overline{z}}{2}$, $\Im(z) = \frac{z-\overline{z}}{2i}$
- $z = re^{i\theta}$: $x = r\cos\theta$, $y = r\sin\theta$
- $|z| = 0 \iff z = 0$
- $\left|z^2\right| = |z|^2 = z\overline{z}$
- $\overline{zw} = (\overline{z})(\overline{w})$
- $1/\overline{z} = 1/\overline{z}$
- $0! = 1$, $f^{(0)} = f$
- $r \downarrow 0$ indicates $r \to 0$ through positive values only
- f is bounded if there exists $M > 0$ s.t. $|f(z)| \le M \forall z \in \mathbb{C}$.

Definition: The Argument

- For $z = re^{i\theta}$, θ is an **argument** of z. $\operatorname{Arg}(z)$ is the unique $\theta \in (-\pi, \pi]$
- $\arg(z) = \{\theta : z = |z|e^{i\theta}\} = \{\operatorname{Arg}(z) + 2k\pi : k \in \mathbb{Z}\}$
- $\arg(zw) = \arg(z) + \arg(w)$, $\arg(\overline{z}) = -\arg(z)$
- $\operatorname{Arg}(zw) = \operatorname{Arg}(z) + \operatorname{Arg}(w) \pm 2k\pi$, $\operatorname{Arg}(\overline{z}) = -\operatorname{Arg}(z) \pm 2k\pi$
- $\operatorname{Arg}_\phi(z)$ is the branch of $\arg(z)$ s.t $\phi < \operatorname{Arg}_\phi(z) \le \phi + 2\pi$
- Any argument θ, ϕ, etc. is always a real number

Theorem: Properties Of Sets $S \subseteq \mathbb{C}$

- **Open** if $\forall\ z_0 \in U$, $\exists \epsilon > 0$ s.t. $D_\epsilon(z_0) \subseteq U$. For example, $D_\epsilon(z), D'_\epsilon(z)$.
- **Closed** if its complement is open, and iff $S = \overline{S}$. For example $\overline{D_\epsilon}(z)$
- A **neighbourhood** of z_0 is an open set containing it.
- z_0 is a **limit point** of S if $\forall \epsilon > 0, D'_\epsilon(z_0) \cap S$ is nonempty
- $\overline{S}$, the **closure** of S, is the union of S with its limit points.
- **Bounded** if $\exists M > 0$ s.t. $|z| \le M$, for all $z \in S$. **Unbounded** otherwise.

Definition: Complex Sequences

Complex sequences $z_n \in \mathbb{C}$ are analogous to real ones, except index from 0.

- $z = \lim_{n\to\infty} z_n$ if $\forall \epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|z_n - z| < \epsilon$ for $n \ge N$.
- z_n can be split into the combination of real sequences, $z_n = a_n + ib_n$. $z = \lim_{n\to\infty} z_n$ iff $\Re(z) = \lim_{n\to\infty} a_n$ and $\Im(z) = \lim_{n\to\infty} b_n$. z_n converges iff a_n and b_n converge.
- Cauchy, Bolzano-Weierstrass, boundedness are all akin to $\mathbb{R}$
- $z \in \overline{S}$ iff there exists a sequence $z_n \in S$ s.t. $z = \lim_{n\to\infty} z_n$

Definition: Complex Functions

$f(z)$ can be written as $f(x+iy) = u(x,y) + iv(x,y)$, where $u,v : \mathbb{R}^2 \to \mathbb{R}$

- For $z_0 \in \overline{S}$, $a_0 = \lim_{z\to z_0} f(z)$ if $\forall\ \epsilon > 0$, $\exists \delta > 0$ s.t. $|f(z) - a_0| < \epsilon$ when $0 < |z - z_0| < \delta$. Standard $\epsilon - \delta$ definition.
- The lim exists iff the lims of the $\Re/\Im$ exist and are the $\Re/\Im$ of the limit
- Let $z_0 \in \overline{S}, f: S \to \mathbb{C}$, $\lim_{z\to z_0} f(z) = a_0$, and $w_n \in S \backslash \{z_0\}$ s.t $w_n \to z_0$, then $\lim_{n\to\infty} f(w_n) = a_0$.
- Limit algebra is analogous to the reals

Definition: Continuity

- $f(z)$ is cont. at z_0 if $\lim_{z\to z_0} f(z)$ exists and equals $f(z_0)$, and cont. on $S \subseteq \mathbb{C}$ if it is cont. at all $z \in S$.
- $f(z)$ is cont. at z_0 iff u,v are cont. at (x_0, y_0).
- $+/\times/\div$(where denominator $\neq 0$) of cont. funcs. preserves continuity.
- f is cont. iff the preimage $f^{-1}(U)$ is open $\forall$ open $U \subseteq \mathbb{C}$
- Continuous functions map clounded sets to clounded sets

Major Definition: Differentiability and Holomorphicity

- $f(z)$ diffable (and thus cont.) at z_0 if $\lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists.
- $f(z)$ is hol. at z_0 if it is diffable on a neighbourhood of z_0
- $+/\cdot/\div$(where denom $\neq 0$) of hol. funcs. preserves hol.city
- Composition preserves diffability, $(f \circ g)'(z_0) = f'(g(z_0))g'(z_0)$
- $\overline{z}$, $|z|$, $|z|^2$, $\Re(z)$, $\Im(z)$ nowhere hol.

Major Theorem: The Cauchy-Riemann Equations

The **Cauchy-Riemann equations** state that if $f(z)$ is diffable at z_0 then $\frac{\partial u}{\partial x}(x_0,y_0) = \frac{\partial v}{\partial y}(x_0,y_0)$ and $\frac{\partial u}{\partial y}(x_0,y_0) = -\frac{\partial v}{\partial x}(x_0,y_0)$

- If u,v are cont.sly diffable (derivative is cont.) on a neighbourhood of (x_0,y_0) and satisfy the C-R eqns at (x_0,y_0), then f is diffable at z_0.
- Functions which are non-constant and purely real-valued or imaginary-valued are nowhere hol.

Definition: Harmonic Functions

$h: \mathbb{R}^2 \to \mathbb{R}$ is **harmonic** if $\forall (x,y) \in \mathbb{R}^2$: $\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$ (Laplace).

- Holomorphicity implies harmonicity
- For a harmonic u, v is a **harmonic conjugate of u** if $f = u + iv$ is hol.

Definition: Polynomials and Rational Functions

- For polynom $P(z) = \sum_{n=0}^{N} a_n z^n$, coeffs $a_i \in \mathbb{R}$, $P(z_0) = 0$ iff $P(\overline{z_0}) = 0$
- For polynoms P, Q, $R := \frac{P(z)}{Q(z)} : \{z \in \mathbb{C} : Q(z) \neq 0\} \to \mathbb{C}$ is a **rational func**.
- The domain of a rational function is always an open set
- Rational functions are holomorphic on $\{z \in \mathbb{C} : Q(z) \neq 0\}$

Major Definition: exp(z)and Complex Trig

- $\exp(z) := e^x(\cos(y) + i\sin(y))$
- $\exp(z + 2\pi i) = \exp(z) = \sum_{j=0}^{\infty} \frac{z^j}{j!}$
- $\exp'(\alpha t) = \alpha \exp(\alpha t)$, $t \in \mathbb{R}, \alpha \in \mathbb{C}$.
- $\sin(z) = \frac{\exp(iz) - \exp(-iz)}{2i}$
- $\cos(z) = \frac{\exp(iz) + \exp(-iz)}{2}$
- $\sin'(z) = \cos(z)$, $\cos'(z) = -\sin(z)$
- $\cos(\theta + \pi) = -\cos(\theta)$ same for sin
- $\cos^2(z) + \sin^2(z) = 1$
- $\tan(z) := \frac{\sin(z)}{\cos(z)}$, $\cot(z) := \frac{1}{\tan(z)}$
- $\sec(z) := \frac{1}{\cos(z)}$, $\csc(z) = \frac{1}{\sin(z)}$
- $\sinh(z) := \frac{\exp(z) - \exp(-z)}{2}$
- $\cosh(z) := \frac{\exp(z) + \exp(-z)}{2}$
- Periodicity, $\frac{\pi}{2}$ offset akin to $\mathbb{R}$
- $\sin(x+iy) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$
- $\cos(x+iy) = \cos(x)\cosh(y) - i\sin(x)\sinh(y)$
- $\sin(z+w) = \sin(z)\cos(w) + \cos(z)\sin(w)$
- $\cos(z+w) = \cos(z)\cos(w) - \sin(z)\sin(w)$
- $\sinh(iz) = i\sin(z)$, $\cosh(iz) = \cos(z)$
- $\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} z^{2n+1} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \ldots$
- $\cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \ldots$

Major Definition: The Complex Logarithm

$\log(z) := \{w \in \mathbb{C} : \exp(w) = z\} = \{\ln|z| + i\phi : \phi \in \arg(z)\}$
$\operatorname{Log}(z) := \ln|z| + i\operatorname{Arg}(z)$

- $\log(re^{i\theta}) = \ln|z| + i\arg(z) = \{\ln r + i\theta + 2k\pi i : k \in \mathbb{Z}\}$
- $\log(zw) = \log(z) + \log(w)$, $\log(\frac{1}{z}) = -\log(z)$
- $\operatorname{Log}_\phi(z) := \ln|z| + i\operatorname{Arg}_\phi(z)$ hol. on D_ϕ, $\operatorname{Log}'_\phi(z) = \frac{1}{z}$ for all $z \in D_\phi$

Definition: Branches and Cuts

Motivated by $\operatorname{Arg}(z)$'s discontinuity on the negative real line, which means $\operatorname{Log}(z)$ is not hol. on all of $\mathbb{C}$.

- A branch cut is a line $L \subseteq \mathbb{C}$ which is such that that a holomorphic branch of a multivalued func may be defined on $\mathbb{C} \backslash L$
- $L_{z_0,\phi} := \{z \in \mathbb{C} : z = z_0 + re^{i\phi}$ for $r > 0\}$. $L_{0,\phi} = L_\phi$.
- $D_{z_0,\phi} = \mathbb{C} \backslash L_{z_0,\phi}$. $D_{0,\phi} = D_\phi$. $D_{-\pi} = D$.
- If $U \subseteq \mathbb{C}$ open, $g: U \to \mathbb{C}$ hol. on U; $\operatorname{Log}_\phi(g(z))$ is hol. on $U \cap g^{-1}(D_\phi)$. If g hol. on $\mathbb{C}$, $\operatorname{Log}_\phi(g)$ is hol. on $g^{-1}(D_\phi) = \{z \in \mathbb{C} : g(z) \in D_\phi\}$.

Definition: Complex Powers

$z^\alpha := \{\exp(\alpha w) : w \in \log(z)\} = \{\exp(\alpha \operatorname{Log}(z))\exp(i\alpha 2k\pi) : k \in \mathbb{Z}\}$

- Let $\alpha, z \in \mathbb{C}, z \neq 0$:
  - (i) If $\alpha \in \mathbb{Z}$ z^α has exactly one value.
  - (ii) If $\alpha = \frac{p}{q}$, $p,q \in \mathbb{Z}, p,q$ coprime, $q \neq 0$, z^α has exactly q values.
  - (iii) If α is irrational or non-real, z^α has infinitely many values.
- If $q \in \mathbb{Z}^+$, $1^{\frac{1}{q}} = 1, w, \ldots, w^{q-1}$, $w = \exp(\frac{2\pi i}{q})$ are the q **roots of unity**
- Each branch of $\log(z)$ gives a branch of z^α.
- The **principal branch** is $z^\alpha = \exp(\alpha \operatorname{Log}(z))$
- For $z \neq 0$, $z^\alpha z^\beta = z^{\alpha+\beta}$ choosing the principal branch for each power.
- A branch of z^α is holomorphic on the cut plane D_ϕ on which Log_ϕ is holomorphic, and $\forall z \in D_\phi$, $\frac{d}{dz} z^\alpha = \alpha z^{\alpha-1}$.

Conformal Maps and Möbius Transforms

Definition: Conformal Map

For open $U \subseteq \mathbb{C}, f: U \to \mathbb{C}$ is **conformal** if the angle between straight lines is equal to the angle between their images under f. This extends to diffable curves by considering their tangents.

- A holomorphic $f(z)$ is conformal for all $z_0 \in U$ where $f'(z_0) \neq 0$

Definition: The Extended Complex Plane

The **extended complex plane** is $\tilde{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$.
Let $a, b \in \mathbb{C}, b \neq 0$, then $a + \infty = \infty$, $b \cdot \infty = \infty$, $\frac{b}{0} = \infty$, $\frac{b}{\infty} = 0$.
Then $f(-\frac{d}{c}) = \infty$ and $f(\infty) = \frac{a}{c}$. We do not define $0 \cdot \infty$ or $\infty + \infty$.

Definition: Möbius Transforms

A **Möbius transformation** is an f s.t. $f(z) = \frac{az+b}{cz+d}$, $a,b,c,d \in \mathbb{C}, ad \neq bc$

- $\frac{\lambda az + \lambda b}{\lambda cz + \lambda d} = \frac{az+b}{cz+d}$, can always set $\lambda = \sqrt{(ad-bc)}$ and impose $ad - bc = 1$
- $f_M(z) = \frac{az+b}{cz+d}$ corresponds to $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\det(M) = ad - bc = 1$.

Translation	Rotation	Dilation	Inversion
$f(z) = z + b$	$f(z) = e^{i\theta} z$	$f(z) = rz, r > 0$	$f(z) = \frac{1}{z}$
$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$	$\begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix}$	$\begin{pmatrix} \sqrt{r} & 0 \\ 0 & 1/\sqrt{r} \end{pmatrix}$	$\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$

- To find the image of a set under an MT - if bdary of set is a circline, bdary of the image is. So calculate image of 3 points on bdary to find bdary of image, and one point inside to know which side of bdary the image is on.
- $(z - (1\text{or } i))/(z + (1\text{or } i))$ maps $\{z \in \mathbb{C} : (\Re(z) \text{ or } \Im(z)) > 0\}$ onto $D_1(0)$

Theorem: Möbius Transform Properties

Let $f(z)$ be a Möbius Transform and $z_2, z_3, z_4 \in \tilde{C}$ be distinct. Then:

- Every MT is a composition of a finite number of translations, rotations, and dilations, and iff f does not fix the point at infinity, one inversion.
- If $c = 0$, $f(\infty = \infty), f(z) = (a/d)z + b/d = f_2 \circ f_1$; $f_1(z) := (a/d)z$ (dilation+rotation), $f_2(z) = z + b/d$ (translation).
- If $c \neq 0$, $f(z) = \frac{(bc-ad)/(c^2)}{z + d/c} + \frac{a}{c} = g_4 \circ g_3 \circ g_2 \circ g_1$; $g_1(z) = z + d/c$ (translation), $g_2(z) = 1/z$ (inversion), $g_3(z) = ((bc-ad)/(c^2))z$ (dilation+rotation), $g_4(z) = z + a/c$ (translation).
- MTs map circlines to circlines
- $f_{M_1 M_2} = f_{M_1} \circ f_{M_2}$ and $f_{M^{-1}} = f_M^{-1}$
- If $f(z_2) = z_2, f(z_3) = z_3, f(z_4) = z_4$. $f = id$.
- There exists a unique f s.t. $f(z_2) = 1, f(z_3) = 0, f(z_4) = \infty$)
- For triplets of distinct points $(z_2, z_3, z_4), (w_2, w_3, w_4) \in \tilde{C}$, $\exists$ a unique MT f s.t. $f(z_2) = w_2, f(z_3) = w_3, f(z_4) = w_4$
- f preserves the cross-ratio: $[f(z_1), f(z_2), f(z_3), f(z_4)] = [z_1, z_2, z_3, z_4]$

Definition: The Riemann Sphere, Stereographic Projection

Describe $\mathbb{R}^3$ with $\{X, Y, Z\}$. $\mathbb{C}$ is the plane $Z = 0$ and $X + iY$ is $\{X, Y, 0\}$:

- The **Riemann Sphere** is $S^2 := \{(X,Y,X) \in \mathbb{R}^3 : X^2 + Y^2 + Z^2 = 1\}$.
- $\phi : \tilde{\mathbb{C}} \to S^2$ is defined by: $\phi(z), z, N$ collinear, and $\phi(\infty) = N := \{0,0,1\}$.
- The **stereographic projection** is $\psi := \phi^{-1}(z) : S^2 \to \mathbb{C}$.
- ψ maps circles to circlines.
- $\phi(z) = (\frac{2x}{|z|^2+1}, \frac{2y}{|z|^2+1}, \frac{|z|^2-1}{|z|^2+1})$. $\psi(X,Y,Z) = \frac{X+iY}{1-Z}$, $\psi(N) = \infty$.

Major Definition: The Cross-Ratio

For distinct $z_1, z_2, z_3, z_4 \in \tilde{C}$,
their **cross-ratio**, $[z_1, z_2, z_3, z_4]$, is the image of z_1 under the unique f s.t. $f(z_2) = 1, f(z_3) = 0, f(z_4) = \infty$. $[z_1, z_2, z_3, z_4] = \frac{z_1 - z_3}{z_1 - z_4} \frac{z_2 - z_4}{z_2 - z_3}$.

$[\infty, z_2, z_3, z_4] = \frac{z_2 - z_4}{z_2 - z_3}$, $[z_1, \infty, z_3, z_4] = \frac{z_1 - z_3}{z_1 - z_4}$,

$[z_1, z_2, \infty, z_4] = \frac{z_2 - z_4}{z_1 - z_4}$, $[z_1, z_2, z_3, \infty] = \frac{z_1 - z_3}{z_2 - z_3}$.

Complex Integration

Definition: Complex Integration Of A Real Variable

For $[a,b] \subseteq \mathbb{R}$, $f: [a,b] \to \mathbb{C}$ is **integrable** if $u,v : [a,b] \to \mathbb{R}$ are int.able. Then $\int_a^b f(t)\ \mathrm{d}t = \int_a^b u(t)\ \mathrm{d}t + i\int_a^b v(t)\ \mathrm{d}t$. Continuous functions are integrable.
This integration respects linearity, FTC, and $\left|\int_a^b f(t)\ \mathrm{d}t\right| \le \int_b^a |f(t)|\ \mathrm{d}t$

Definition: Contours And Their Types

For $z_0, z_1 \in \mathbb{C}$, a **(parametrised) curve** Γ connecting them is a continuous func $\gamma : [t_0, t_1] \to \mathbb{C}$ for $t_0, t_1 \in \mathbb{R}, t_0 < t_1$, s.t. $\gamma(t_0) = z_0$ and $\gamma(t_1) = z_1$.

- γ can be split into $\Re/\Im$ parts by two continuous real functions $x, y : [t_0, t_1] \to \mathbb{R}$ s.t. $\gamma(t) = x(t) + iy(t)$ for all $t \in [t_0, t_1]$
- Γ is **regular** if γ is continuously diffable and $\gamma'(t) \neq 0$ for all $t \in (t_0, t_1)$
- Γ is a closed and bounded subset of $\mathbb{C}$
- For regular Γ, **arclength** $\ell(\Gamma) := \int_{t_0}^{t_1} \left|\gamma'(t)\right| \mathrm{d}t = \int_{t_0}^{t_1} \sqrt{x'(t)^2 + y'(t)^2}\ \mathrm{d}t$
- For Γ an arc of a circle of radius r through an angle θ, $\ell(\Gamma) = r\theta$
- A curve Γ is a **contour** if it is a finite union of regular curves $\Gamma_1, \ldots, \Gamma_N$ s.t $\gamma_n(t_1) = \gamma_{n+1}(t_0)$, ie the endpoints join up.
- A contour is **simple** if it self-intersects nowhere, except possibly at t_0/t_1.
- A **loop** is a simple closed contour
- A loop Γ is **positively-oriented** if, going around Γ, $\operatorname{Int}(\Gamma)$ is on the left.

Major Definition: Contour Integrals

Let $z_0, z_1 \in \mathbb{C}$, Γ be a contour from z_0 to z_1, and $f : \Gamma \to \mathbb{C}$ be cont. Then the **integral of f along** Γ is defined by $\int_\Gamma f(z)\ \mathrm{d}z = \int_{t_0}^{t_1} f(\gamma(t))\gamma'(t)\ \mathrm{d}t$

- Contour integration is linear
- For two parametrisations of Γ, $\gamma : [t_0, t_1] \to \mathbb{C}$ and $\tilde{\gamma} : [\tilde{t_0}, \tilde{t_1}] \to \mathbb{C}$, if there exists injective diffable $\lambda : [\tilde{t_0}, \tilde{t_1}] \to [t_0, t_1]$ with $\lambda'(t) > 0 \forall t \infty [\tilde{t_0}, \tilde{t_1}]$ s.t. $\tilde{\gamma}(t) = \gamma(\lambda(t))\ \forall\ t \in [\tilde{t_0}, \tilde{t_1}]$, then the integrals along γ and $\tilde{\gamma}$ are equal.
- If Γ a regular curve, $f : \Gamma \to \mathbb{C}$ cont., integrating along the path of a parametrisation of Γ backwards gives the negative of the integral along Γ.
- A contour is **closed** if $\gamma(t_0) = \gamma(t_1)$
- A **contour integral** of f is the sum of the integrals of f along the regular curves which make up the contour.

Definition: Domains

$D \subseteq \mathbb{C}$ is a **domain** if D is open and every two points in D can be connected by a contour lying wholly in D.

- If $u : D \to \mathbb{R}$ is diffable, $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0$ on D, then u is constant on D.
- A cont. f on D has an **antiderivative** F on D if $F'(z) = f(z)\ \forall z \in D$.
- The integral of cont. f with an antiderivative on D along a closed Γ in D is 0.
- D is **simply-connected** if $\operatorname{Int}(\Gamma) \subseteq D$ for all loops $\Gamma \subseteq D$.

Lemma: The M-L Lemma

Let Γ be a regular curve in $\mathbb{C}$ and $f : \Gamma \to \mathbb{C}$ be continuous.
Then $\left|\int_\Gamma f(z)\ \mathrm{d}z\right| \le \max_{z \in \Gamma} |f(z)| \ell(\Gamma)$

Lemma: Path-Independence

For a domain $D \subseteq C$ and cont. $f : D \to \mathbb{C}$, the following are equivalent:

- (i) f has an antiderivative F on D (thus is cont.)
- (ii) $\int_\Gamma f(z)\ \mathrm{d}z = 0$ for all closed Γ in D
- (iii) All contour integrals $\int_\Gamma f(z)\ \mathrm{d}z$ are independent of the path Γ, and depend only on the endpoints of the contour.

Theorem: Jordan Curve Theorem

For $\Gamma \subseteq \mathbb{C}$ a loop, Γ partitions $\mathbb{C}$ into two domains: The bdd **interior** of Γ, $\operatorname{Int}(\Gamma)$; and the unbdd **exterior** of Γ, $\operatorname{Ext}(\Gamma)$. $\mathbb{C} = \operatorname{Int}(\Gamma) \cup \Gamma \cup \operatorname{Ext}(\Gamma)$

Major Theorem: Cauchy Integral Theorem

Let Γ be a loop and f be hol. inside and on Γ. Then $\int_\Gamma f(z)\ \mathrm{d}z = 0$.

- If f is hol. on a simply-connected domain D, f has an antiderivative on D
- If $z_0 \in \mathbb{C}$ and $z_0 \notin \Gamma$, then $\int_\Gamma \frac{1}{z - z_0}\ \mathrm{d}z = 2\pi i$ if $z \in \operatorname{Int}(\Gamma)$ and 0 otherwise

Theorem: The Deformation Theorem

Let $\Gamma_1, \Gamma_2 \subseteq \mathbb{C}$ be loops and f hol. on: $\Gamma_1; \Gamma_2; \{\operatorname{Int}(\Gamma_1) \backslash \operatorname{Int}(\Gamma_2)\} \cup \{\operatorname{Int}(\Gamma_2) \backslash \operatorname{Int}(\Gamma_1)\}$. Then $\int_{\Gamma_1} f(z)\ \mathrm{d}z = \int_{\Gamma_2} f(z)\ \mathrm{d}z$

Major Theorem: Cauchy Integral Formula, Generalised CIF

Let Γ be a loop, $z_0 \in \operatorname{Int}(\Gamma)$, and f hol. inside and on Γ. Then:

- $f(z_0) = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{z - z_0}\ \mathrm{d}z$
- f is ∞'ly diffable at z_0, for $n \in \mathbb{N}$, $f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\Gamma \frac{f(z)}{(z - z_0)^{n+1}}\ \mathrm{d}z$

- Can be rearranged to evaluate contour integrals given appropriate $f^{(n)}$
- For $z_0 \in$ domain D, $R > 0$ s.t. $\overline{D}_R(z_0) \subseteq D$, f hol. on D, and $M > 0$ an upper bound for $|f(z)|$, then for all $n \in \mathbb{N}$, $\left|f^{(n)}(z_0)\right| \le \frac{n! M}{R^n}$

Theorem: Morera's Theorem

Let $D \subseteq \mathbb{C}$ be a domain, $f : D \to \mathbb{C}$ continuous, and $\int_\Gamma f(z)\ \mathrm{d}z = 0$ for all loops Γ inside D. Then f is hol. on D.

Theorem: Liouville's Theorem

Let f be hol. on $\mathbb{C}$ and bounded. Then f is constant.

- For $D \subseteq \mathbb{C}$ a domain, $z_0 \in D$, $R > 0$ s.t. $\overline{D}_R(z_0) \subseteq D$, and f hol. on D; $f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + R\exp(it))\ \mathrm{d}t$.
- If f s.t. $\max_{z \in \overline{D}_R(z_0)} |f(z)| = |f(z_0)|$, $f(z)$ is constant on $\overline{D}_R(z_0)$.

Theorem: The Maximum Modulus Principle

Let $D \subseteq \mathbb{C}$ be a domain, and f be hol. and bounded on D.
Then if $|f(z)|$ achieves its maximum on D, f is constant on D.

- If a func is hol. on a bounded domain, and cont. up to and including the boundary, it attains the maximum of its modulus on the boundary.

Thm: Maximum/Minimum Principle For Harmonic Funcs

Let $D \subseteq \mathbb{R}^2$ be a domain, $\phi : D \to \mathbb{R}$ be harmonic s.t. ϕ is bounded above or below on D by $M > 0$, and $\phi(z_0) = M$ for some $z_0 \in D$.
Then ϕ is constant on D.

Example: Maximum Size Of a Polynomial On A Disc

Find the maximum value of $\left|z^2 + 3z - 1\right|$ on the unit disc $\overline{D}_1(0)$.
$z^2 + 3z - 1$ is hol. on $\mathbb{C}$ so its max occurs somewhere on $C_1(0)$. For $|z| = 1$:

$$\begin{aligned}
\left|z^2 + 3z - 1\right|^2 &= \left(z^2 + 3z - 1\right)\left(\overline{z}^2 + 3\overline{z} - 1\right) \\
&= z^2\overline{z}^2 + 3z^2\overline{z} - z^2 + 3z\overline{z}^2 + 9z\overline{z} - 3z - \overline{z}^2 - 3\overline{z} + 1 \\
&= (z\overline{z})^2 + 3(z\overline{z})z - z^2 + 3(z\overline{z})\overline{z} + 9(z\overline{z}) - 3z - \overline{z}^2 - 3\overline{z} + 1 \\
&= 1 + 3z - z^2 + 3\overline{z} + 9 - 3z - \overline{z}^2 - 3\overline{z} + 1 \\
&= 11 - (z^2 + \overline{z}^2) = 11 - 2\Re(z^2)
\end{aligned}$$

Since $\left|z^2\right| = 1, \Re(z^2) \ge 1$ with a min at $z^2 = -1$, where $\left|z^2 + 3z - 1\right|^2 = 13$.
Thus the max value of $\left|z^2 + 3z - 1\right|$ is $\sqrt{13}$, attained at $\pm i$.

Ex.: Finding A Laurent Series On Punctured Disc, Annulus

Let $f = 1/(z(z+i))$. (i) Find the LSE for f valid on $D'_1(0)$.
(ii) Find the LSE for f valid on $A_{1,\infty}(0)$.
(i): For $|z| < 1$;
$\frac{1}{z+i} = \frac{1}{i} \cdot \frac{1}{1 - iz} = -i\sum_{j=0}^{\infty} (iz)^j = \sum_{j=0}^{\infty} -i^{j+1} z^j$, so for $z \in D'_1(0)$;
$f(z) = \frac{1}{z} \cdot \frac{1}{z+i} = \frac{1}{z}\sum_{j=0}^{\infty} -i^{j+1} z^j = \sum_{j=0}^{\infty} i^{j-1} z^{j-1} = \sum_{j=-1}^{\infty} (iz)^j$.
(ii): For $|z| > 1$; $\frac{1}{z+i} = \frac{1}{z} \cdot \frac{1}{1 - (-i/z)} = \frac{1}{z}\sum_{j=0}^{\infty} (-i)^j / z^j = \sum_{j=0}^{\infty} (-i)^j z^{-j-1}$ so for $z \in A_{1,\infty}$; $f(z) = \frac{1}{z} \cdot \frac{1}{z+i} = \frac{1}{z}\sum_{j=0}^{\infty} (-i)^j z^{-j-1} = \sum_{j=0}^{\infty} (-i)^j z^{-j-2} = \sum_{j=-\infty}^{0} (-i)^{-j} z^{j-2} = \sum_{j=-\infty}^{0} i^j z^{j-2} = \sum_{j=-\infty}^{-2} i^{j+2} z^j = \sum_{j=-\infty}^{-2} -i^j z^j$

Example: Exact Values Of A Complex Power

Give all the values of $(1+i)^{1/4}$

$$\begin{aligned}
(1+i)^{1/4} &= \left\{\exp\left(\frac{1}{4}(\ln|1+i| + i\operatorname{Arg}(1+i) + 2k\pi i)\right) : k \in \mathbb{Z}\right\} \\
&= \left\{\exp\left(\frac{\ln\sqrt{2}}{4} + \frac{i\pi}{4} + \frac{ik\pi}{2}\right) : k \in \mathbb{Z}\right\} \\
&= \left\{\exp\left(\ln 2^{1/8} + \frac{i\pi}{16} + \frac{ik\pi}{2}\right) : k \in \mathbb{Z}\right\} = \left\{2^{1/8} e^{i\pi/16} e^{ik\pi/2} : k \in \mathbb{Z}\right\} \\
&= \left\{2^{1/8} e^{i\pi/16}, 2^{1/8} e^{9i\pi/16}, 2^{1/8} e^{17i\pi/16}, 2^{1/8} e^{25i\pi/16}\right\}
\end{aligned}$$

Series Expansions For Holomorphic Functions

Definition: Convergence Of An Infinite Series

An infinite series $\sum_{j=0}^{\infty} z_j$ **converges**, with $\sum_{j=0}^{\infty} z_j = S$, if the sequence $S_n := \sum_{j=0}^{n} z_j$ is convergent with limit $S \in \mathbb{C}$. Otherwise **divergent**.

- If $\sum_{j=0}^{\infty} z_j$ is convergent, $\lim n \to \infty z_n = 0$
- Comparison Test: If z_n is s.t. $|z_n| \le M_n \ge 0$, for all $n \ge N$ for some $N \in \mathbb{N}$, and $\sum_{j=0}^{\infty} M_j$ converges, then $\sum_{j=0}^{\infty} z_j$ converges.
- Geometric Series: For $c \in \mathbb{C}$, $\sum_{j=0}^{\infty} c^j$ converges (to $\frac{1}{1-c}$) iff $|c| < 1$.
- Ratio Test: If $\lim_{n\to\infty} \left| \frac{z_{n+1}}{z_n} \right| = L$, $\sum_{j=0}^{\infty} z_j$ converges if $L < 1$ and diverges if $L > 1$. If $L = 1$ the test gives no information..

Definition: Pointwise And Uniform Convergence

For $S \subseteq \mathbb{C}$ and $f_n : S \to \mathbb{C}$ a sequence of functions:
f_n converge **pointwise** to $f : S \to \mathbb{C}$ if for all $z \in S$ and all $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(z) - f(z)| < \epsilon$ when $n \ge N$. Note N may depend on z.
f_n converge **uniformly** to $f : S \to \mathbb{C}$ if for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $|f_n(z) - f(z)| < \epsilon$ when $n \ge N$. Note N may not depend on z.

- Holo. f_n may converge pointwise to an f which is not even cont.
- If cont. f_n converge uniformly to f, f is cont.
- $\sum_{j=0}^{\infty} f_j(z)$ cnvges ptwise/unifly if $S_n = \sum_{j=0}^{n} f_j$ cnvges ptwise/unifly.
- Weierstrass M-Test: For $S \subseteq C, M_n \ge 0$ s.t. $\forall z \in S$ and $n \ge N \in \mathbb{N}$, $|f_n(z)| \le M_n$, and $\sum_{j=0}^{\infty} M_j$ cnvgent; $\sum_{j=0}^{\infty} f_j(z)$ is unifly cnvgent.
- For $f_n : S \to \mathbb{C}$ unifly cnvgng to f, and $\Gamma \subseteq S$; $\int_\Gamma f_n(z)\ dz \to \int_\Gamma f(z)\ dz$.
- For continuous f_n s.t. $\sum_{j=0}^{\infty} f_j(z)$ converges uniformly on S and $\Gamma \subseteq S$: $\int_\Gamma \sum_{j=0}^{\infty} f_j(z)\ dz = \sum_{j=0}^{\infty} \int_\Gamma f_j(z)\ dz$
- If f_n, hol. on simply-connected D, cnvge unifly to f: f is hol. on D.

Definition: Power Series

For $a_n \in \mathbb{C}$ a complex sequence, a **power series** with **coefficients** a_j is an infinite series of the form $\sum_{j=0}^{\infty} a_j (z - z_0)^j$.
Given a PS, $\exists R \in [0, \infty) \cup \{\infty\}$, the **radius of convergence**, such that:
(i) The series converges on $D_R(z_0)$
(ii) The series converges uniformly on $\overline{D}_r(z_0)$ for $r \in [0, R]$
(iii) The series diverges on $\mathbb{C} \backslash \overline{D}_R(z_0)$

- For $f = \sum_{j=0}^{\infty} a_j (z - z_0)^j$, f is hol. on $D_R(z_0)$
- If $\lim_{j\to\infty} \left| \frac{a_j}{a_{j+1}} \right|$ exists, it is equal to R.

Definition: Taylor Series

For f hol. at $z_0 \in \mathbb{C}$:
the **Taylor series of f centred at** z_0 is the PS $\sum_{j=0}^{\infty} \frac{f^{(j)}(z_0)}{j!} (z - z_0)^j$.

- For f hol. on $D_R(z_0)$, the TS of f centred at z_0 converges to $f(z)$ for all $z \in D_R(z_0)$, uniformly on $\overline{D}_r(z_0)$ for $0 \le r < R$
- Thus the TS of f centred at z_0 converges to f everywhere inside the largest open disc centred at z_0 on which f is hol.
- For U open, f is **analytic** if for all $z \in U$, f can be expressed as a convergent power series centred at z, valid on some disc centred at z.
- If U is open and f hol. , f is analytic
- For f hol. on and $z \in D_R(z_0)$, $f'(z) = \sum_{j=0}^{\infty} \frac{f^{(j+1)}(z_0)}{j!} (z - z_0)^j$
- Convergent PSs can be differentiated termwise
- For f, g hol. on and $z \in D_R(z_0)$:
  TSs respect linearity, and the TS for fg, valid on the same $D_R(z_0)$, is $\sum_{j=0}^{\infty} \frac{1}{j!} \left(\sum_{k=0}^{j} \binom{j}{k} f^{(k)}(z_0) g^{(j-k)}(z_0) \right) (z - z_0)^j$
- Uniqueness: If $\exists$ a PS converging to f on a disc, it is the TS of f.

Definition: Laurent Series

For $z_0 \in \mathbb{C}$ and $\ldots, a_{-1}, a_0, a_1, \ldots \in C$
a doubly-infinite sequence, a **Laurent series** centred at z_0 is a PS of form $\sum_{j=-\infty}^{\infty} a_j (z - z_0)^j = \sum_{j=0}^{\infty} a_j (z - z_0)^j + \sum_{j=1}^{\infty} a_{-j} (z - z_0)^{-j}$

- An LS cnvges if its component PSs converge. The first has RoC R, view the second as a PS in $w = (z - z_0)^{-1}$, RoC $S > 0$. Then the LS cnvges for z s.t. $|z - z_0| < R$ and $\left| (z - z_0)^{-1} \right| < S$, or $S^{-1} = r < |z - z_0| < R$.
- For f hol. on $A_{r,R}(z_0), 0 \le r < R \le \infty$; f can be written as a LS centred at z_0 cnvging on $A_{r,R}(z_0)$, unifly on $\overline{A}_{r',R'}(z_0)$, $r < r' \le R' < R$, $a_j = \frac{1}{2\pi i} \int_\Gamma \frac{f(z)}{(z - z_0)^{j+1}}\ dz$ for Γ in $A_{r,R}(z_0)$ s.t. $z_0 \in \mathrm{Int}(\Gamma)$
- If f is holomorphic on the 'hole' of the annulus, the LS is the TS
- Uniqueness: If the PS $\sum_{j=-\infty}^{\infty} c_j (z - z_0)^j$ cnvgs on $A_{r,R}(z_0)$, $\exists f$ hol. on $A_{r,R}(z_0)$ s.t. its LS centred at z_0 is the PS. Ie, if a func can be expressed as a PS of the form of an LS centred at some z_0, that PS is the LS

Definition: Zeroes

Let $z_0 \in \mathbb{C}$, U a neighbourhood of z_0, f hol. on U. Then:

- z_0 is a **zero** of f if $f(z_0) = 0$.
- z_0 is of **finite order** if there exists $m \in \mathbb{N}$ s.t. $f(z_0) = f'(z_0) = \ldots = f^{(m-1)}(z_0) = 0$ but $f^m(z_0) \ne 0$. Then z_0 is a **zero of order m**.
- A zero of order 1 is a **simple zero**.
- A zero is **isolated** if $\exists R > 0$ s.t. $f(z) \ne 0$ for $z \in D'_R(z_0)$
- If f has a zero of finite order at z_0, then it is isolated
- If f is such that $f(z_n) = 0$ for a sequence of distinct $z_n \in U$ s.t. $z_n \to z_0$ then f is identically zero on some disc centred at z_0. (Proof by Taylor expansion since zero at z_0 is of infinite order, and non-isolated)

Definition: Singularities And Their Classifications

Let D be a domain, $f : D \to \mathbb{C}$. Then:
$z_0 \in \mathbb{C}$ is a **singularity** of f if f is not hol. at z_0.
If z_0 is an isolated singularity of f hol. on $D'_R(z_0)$ for some $R > 0$, then f has an LE $f(z) = \sum_{j=-\infty}^{\infty} a_j (z - z_0)^j$ valid on $A_{0,R}$.

- A singularity z_0 is **removable** if $a_j = 0\ \forall j < 0$ (LE is $\sum_{j=0}^{\infty} a_j (z - z_0)^j$)
- z_0 is a **pole of order m** $\in \mathbb{N}$ if $a_j = 0$ for $j < -m$ and $a_{-m} \ne 0$ (LE is $\sum_{j=-m}^{\infty} a_j (z - z_0)^j$). Order 1/2/3 poles are **simple/double/triple**.
- A singularity z_0 is **essential** if $a_j \ne 0$ for infinitely many neg. values of j

If z_0 is a removable singularity of f hol. on $D'_R(z_0)$ for $R > 0$, then $f(z_0)$ can be redefined s.t. f is hol. at z_0. By definition f has an LE valid on $D'_R(z_0)$, so setting $f(z_0) = a_0$, the value of the LE at z_0, makes it hol.

- A singularity is **isolated** if there exists $R > 0$ s.t. f is hol. on $D'_R(z_0)$.
- If z_0 is a singularity of a rational function $f = \frac{P}{Q}$, it is isolated
- For f, g hol. at z_0, with z_0 a zero of order m of g:
  (i) If z_0 is not a zero of f, then f/g has a pole of order m at z_0
  (ii) If z_0 is a zero of order k of f, then f/g has a pole of order $m - k$ at z_0 if $m > k$, and a removable singularity at z_0 otherwise.

Definition: Analytic Continuation

For $D \subseteq \tilde{D} \subseteq \mathbb{C}$ domains, and $f : D \to \mathbb{C}$ hol., we say a hol. func $F : \tilde{D} \to \mathbb{C}$ is an **analytic continuation** of f if $F(z) = f(z)$ for $z \in D$.

Major Theorem: The Identity Theorem

Let $D \subseteq \mathbb{C}$ be a domain, $z_0 \in D$, f hol. on D, and $f(z) = 0$ for all $z \in D_R(z_0)$. Then $f(z) = 0$ for all $z \in D$.

- If $g : D \to \mathbb{C}$ also hol. s.t. $f(z) = g(z)\ \forall z \in D_R(z_0)$; $f(z) = g(z)\ \forall z \in D$.
- If f s.t. $f(z_n) = 0$ for seq. of distinct $z_n \in D, z_n \to z_0$; $f(z) = 0\ \forall z \in D$.
- If $g(z)$ also s.t. $g(z_n) = 0$ for the same z_n, then $f(z) = g(z)$ for all $z \in D$.

Residue Calculus And Its Applications

Theorem: 5.1.1

Let $z_0 \in \mathbb{C}$, f be hol. on $D'_R(z_0)$ with an isolated singularity at z_0, and Γ a loop inside $D'_R(z_0)$ with $z_0 \in \mathrm{Int}(\Gamma)$. Then $\int_\Gamma f(z)\ dz = 2\pi i a_{-1}$, where a_{-1} is the $j = -1$ coeff in the LE of f centred at z_0 valid on $D'_R(z_0)$.

Definition: The Residue

Let $z_0 \in \mathbb{C}$, f hol. on $D'_R(z_0)$ with an isolated singularity at z_0.
Then the **residue of f at** z_0 is $\mathrm{Res}(f, z_0) = a_{-1}$
where the LE of f valid on $D'_R(z_0)$ is $f(z) = \sum_{j=-\infty}^{\infty} a_j (z - z_0)^j$

Lemma: Residue Calculation Lemmas

For all, let $z_0 \in \mathbb{C}$ and f hol. on $D'_R(z_0)$

- 5.1.4: If f has a removable singularity at z_0, $\mathrm{Res}(f, z_0) = 0$.
- 5.1.5: If f has a pole of order m at z_0 then $\mathrm{Res}(f, z_0) = \lim_{z \to z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} ((z - z_0)^m f(z))$
- 5.1.7: If $f = g/h$, g, h hol. on $D_R(z_0)$ s.t. h has a simple zero at z_0 and $g(z_0) \ne 0$, then $\mathrm{Res}(f, z_0) = \frac{g(z_0)}{h'(z_0)}$

Major Theorem: The Cauchy Residue Theorem

Let Γ be a loop, and f hol. inside and on Γ except for finitely many isolated singularities $z_1, \ldots, z_k \in \mathrm{Int}(\Gamma)$.
Then $\int_\Gamma f(z)\ dz = 2\pi i \sum_{j=1}^{k} \mathrm{Res}(f, z_j)$

Definition: Meromorphicity

Let $D \subseteq \mathbb{C}$ be a domain. Then: f is **meromorphic on** D if for all $z \in D$, f has a pole of finite order at z or f is hol. at z.

- If Γ a loop in D, f mero. on D and not identically 0, then f has a finite number of poles and zeroes on $\mathrm{Int}(\Gamma)$.

Definition: $N_0(f)$ and $N_\infty(f)$

Let Γ be a loop and f mero. on $\mathrm{Int}(\Gamma)$, with zeroes $w_1, \ldots, w_l$ and poles $z_1, \ldots, z_k$ all $\in \mathrm{Int}(\Gamma)$.
Then $N_0(f)$, the number of zeroes of f inside Γ, and $N_\infty(f)$, the number of poles of f inside Γ, both counted with multiplicity, are:
$N_0(f) = \sum_{j=1}^{l} \text{order of } w_j$, and $n_\infty(f) = \sum_{j=1}^{k} \text{order of } z_j$.

Theorem: The Argument Principle

Let Γ be a loop, f be mero. on $\mathrm{Int}(\Gamma)$, and hol. and non-zero on Γ. Then: $\int_\Gamma \frac{f'(z)}{f(z)}\ dz = 2\pi i (N_0(f) - N_\infty(z))$.

- If f is hol. inside and on Γ, then $\int_\Gamma \frac{f'(z)}{f(z)}\ dz = 2\pi i (N_0(f))$

Theorem: Rouché's Theorem

Let Γ be a loop and f, g hol. on and inside Γ s.t. for all $z \in \Gamma$, $|f(z) - g(z)| < |f(z)|$. Then $N_0(f) = N_0(g)$.

- FTA: A degree n polynom has exactly n zeros counted with multiplicity

Lemma: 5.6.5 - The Binomial As An Integral

Let $0 \le k \le n$ be non-negative integers, Γ a loop with $0 \in \mathrm{Int}(\Gamma)$, and $\binom{n}{k} = \frac{n!}{k!(n-k!)}$. Then $\binom{n}{k} = \frac{1}{2\pi i} \int_\Gamma \frac{(1+z)^n}{z^{k+1}}\ dz$.

Major Theorem: Open Mapping Theorem

Let $D \subseteq \mathbb{C}$ be a domain, and f non-constant and hol. on D. Then the image of D under f, $f(D) = \{f(z) : z \in D\}$ is an open subset of $\mathbb{C}$

- Let f be hol. on D s.t. any of $\Re(f(z)), \Im(f(z)), |f(z)|, \mathrm{Arg}(f(z))$ are constant. Then f is constant.
- Schwarz Lemma: Let f be hol. on $D_1(0)$ s.t. $f(0) = 0$ and $|f(z)| \le 1$ for all $z \in D_1(0)$. Then $|f(z)| \le |z|$ for all $z \in D_1(0)$.

Definition: Improper Integrals

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. Then
(i) $\int_0^{\infty} f(x)\ dx := \lim_{R\to\infty} \int_0^R f(x)\ dx$
(ii) $\int_{-\infty}^{0} f(x)\ dx := \lim_{R\to-\infty} \int_R^0 f(x)\ dx$
(iii) $\int_{-\infty}^{\infty} f(x)\ dx := \lim_{R_1, R_2 \to (\infty, -\infty)} \int_{R_2}^{R_1} f(x)\ dx$
whenever the appropriate limits exist.
If $\int_{-\infty}^{\infty} f(x)\ dx$ exists, it is equal to $\lim_{R\to\infty} \int_R^R f(x)\ dx$.
Let $c \in (a, b) \subseteq \mathbb{R}$ and $f : [a, b] \backslash \{c\} \to \mathbb{R}$ be cont.. Then:
(i) $\int_a^c f(x)\ dx := \lim_{r \downarrow 0} \int_a^{c-r} f(x)\ dx$
(ii) $\int_c^b f(x)\ dx := \lim_{s \downarrow 0} \int_{c+s}^{b} f(x)\ dx$
(iii) $\int_a^b f(x)\ dx := \lim_{r \downarrow 0} \int_a^{c-r} f(x)\ dx + \lim_{s \downarrow 0} \int_{c+s}^{b} f(x)\ dx$
whenever the appropriate limits exist. If the limits in the third case exist the result is equal to the p.v. of the integral.

Definition: Principal Value

For $c \in \mathbb{R}$, $f : \mathbb{R} \backslash \{c\} \to \mathbb{R}$ cont., the **principal value** of the integral is:
$\mathrm{p.v.} \int_{-\infty}^{\infty} f(x)\ dx = \lim_{R\to\infty, r \downarrow 0} \left(\int_{-R}^{c-r} f(x)\ dx + \int_{c+r}^{R} f(x)\ dx \right)$
provided the two limits exist independently

- If $\lim_{R\to\infty} \int_R^R f(x)\ dx$ exists and $\int_{-\infty}^{\infty} f(x)\ dx$ does not, define the **Cauchy principal value** by $\mathrm{p.v.} \int_{-\infty}^{\infty} f(x)\ dx := \lim_{R\to\infty} \int_R^R f(x)\ dx$.

Lemma: Jordan Lemma

Let P, Q be polynoms with $\deg(Q) \ge \deg(P) + 1$, $a \in \mathbb{R} \ne 0$.
Then $\lim_{R\to\infty} \int_{C_R^+} \exp(iaz) \frac{P(z)}{Q(z)}\ dz = 0$ if $a > 0$
and $\lim_{R\to\infty} \int_{C_R^-} \exp(iaz) \frac{P(z)}{Q(z)}\ dz = 0$ if $a < 0$.

Lemma: 5.5.3 - Integrating On An Arc

Let $D \subseteq C$ be a domain, $c \in D$, f mero. on D with a simple pole at $z \in D$, and S_r be the circular arc paremed. by $\gamma(\theta) = c + r\exp(i\theta)$ for $\theta \in [\theta_0, \theta_1], 0 \le \theta_0 < \theta_1 \le 2\pi$. Then $\lim_{r \downarrow 0} \int_{S_r} f(z)\ dz = i(\theta_1 - \theta_0) \mathrm{Res}(f, c)$

Examples

Example: Finding A Holomorphic Branch

Find a branch of $f(z) = (z^2 - 9)^{1/4}$ s.t. $f(z)$ is hol. at 0. $f(z) = \exp(\frac{1}{4} \log(z^2 - 9))$, so must find branch of $\log(z^2 - 9)$ hol. at 0.
$(z^2 - 9) = (z - 3)(z + 3)$, so $(\log(z - 3) + \log(z + 3)) = \log(z^2 - 9)$.
$\mathrm{Log}(z - 3)$ is hol. on $\{z \in \mathbb{C} : z - 3 \in D_{\phi_1}\} = \{z \in \mathbb{C} : z \in D_{3,\phi_1}\}$, so choose any ϕ_1, for example $\phi_1 = 0$, s.t. $0 \in D_{3,\phi_1}$
$\mathrm{Log}_{\phi_2}(z + 3)$ is hol. on $\{z \in \mathbb{C} : z + 3 \in D_{\phi_2}\} = \{z \in \mathbb{C} : z \in D_{-3,\phi_2}\}$, so choose any ϕ_2, for example $\phi_2 = -\pi$, s.t. $0 \in D_{-3,\phi_2}$.
Then $g(z) = \mathrm{Log}_0(z - 3) + \mathrm{Log}(z + 3)$ is a branch of $\log(z^2 - 9)$ hol. at 0, and our branch is $f(z) = \exp(\frac{1}{4} \mathrm{Log}_0(z - 3) + \mathrm{Log}(z + 3))$.

Example: Finding Zeroes Of A Polynomial

Prove $h(z) = 1/(z^4 + z + 4)$ is hol. on $D_1(0)$ but not on $D_2(0)$.
Show $p(z) = z^4 + z + 4$ has no zeroes in $D_1(0)$ and one or more in $D_2(0)$.
Let $q_1(z) := 4$. Then on $C_1(0)$; $|p(z) - q_1(z)| = \left| z^4 + z + 4 - 4 \right| = \left| z^4 + z \right| \le \left| z^4 \right| + |z| = 1^4 + 1 = 2 < 4 = q_1(z)$.
By Rouché's Theorem, $N_0(p) = N_0(q_1) = 0$ on $D_1(0)$, so h hol on $D_1(0)$.
Let $q_2(z) = z^4$. Then on $C_2(0)$; $|p(z) - q_2(z)| = \left| z^4 + z + 4 - z^4 \right| = |z + 4| \le |z| + |4| = 6 < 16 = \left| z^4 \right| = |q_2(z)|$.
Again by Rouché, $N_0(p) = N_0(q_2) = 4$ on $D_2(0)$, so h not hol. on $D_2(0)$.

Example: Trigonometric Integrals

Integrals of form $\int_0^{2\pi} R(\cos\theta, \sin\theta)\ d\theta$ for rational R can be evaluated by considering a contour integral of an appropriate function around $C_1(0)$. If $z = e^{i\theta} \in C_1(0)$, $\overline{z} = \frac{1}{z}$, $\cos\theta = \Re(z) = \frac{z + (1/z)}{2}$, $\sin\theta = \Im(z) = \frac{z - (1/z)}{2i}$
If $f(z) := \frac{1}{iz} R\left(\frac{z + 1/z}{2}, \frac{z - 1/z}{2i} \right)$; $f(\exp(i\theta)) = \frac{1}{i\exp(i\theta)} R(\cos\theta, \sin\theta)$.
Parametrise $C_1(0)$ by $\gamma : [0, 2\pi] \to \mathbb{C}, \gamma(\theta) = \exp(i\theta)$. Then
$\int_{C_1(0)} f(z)\ dz = \int_0^{2\pi} f(\gamma(\theta)) \gamma'(\theta)\ d\theta = \int_0^{2\pi} f(\exp(i\theta)) i\exp(i\theta)\ d\theta = \int_0^{2\pi} \frac{1}{i\exp(i\theta)} R(\cos\theta, \sin\theta) i\exp(i\theta)\ d\theta = \int_0^{2\pi} R(\cos\theta, \sin\theta)\ d\theta$. So we may transform a real integral to a contour integral, then apply the CRT.

Example: Evaluating Infinite Series With Contour Integrals

Consider the alternating sum $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.
Let $f = \csc(\pi z)/z^2 = \frac{1}{\sin(\pi(z))} \frac{1}{z^2}$ which is hol. on $\mathbb{C}$ except for poles at $z = n \in \mathbb{Z}$. For non-zero n, the pole is simple, and we can calculate that $\mathrm{Res}(f, n) = \frac{1}{\pi\sin'(\pi z)} \frac{1}{n^2} = \frac{1}{\pi\cos(\pi n)} \frac{1}{n^2} = (-1)^n \frac{1}{\pi n^2}$,
To find $\mathrm{Res}(f, 0)$ we compute the LE centred at 0, remembering that we only care about the a_1 term so can ignore most higher order terms:

$$f(z) = \frac{1}{z^2} \frac{1}{\sin(\pi z)}$$
$$= \frac{1}{z^3} \frac{1}{1 - (\pi z)^2/3! + \ldots}$$
$$= \frac{1}{\pi z^3} \frac{1}{(1 - (\pi z)^2/6 - \ldots)}$$
$$= \frac{1}{\pi z^3} (1 + ((\pi z)^2)/3! + \ldots)$$
$$+ \left(\frac{(\pi z)^2}{6} + \ldots \right)^2 + \ldots) = \frac{1}{\pi z^3} + \frac{\pi}{6z} + \ldots$$

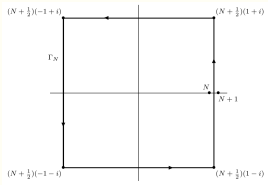

FIGURE 10. The contour Γ_N.

Thus $\mathrm{Res}(f, 0) = \pi/6$. Now consider Γ_N as shown, for $N \in \mathbb{N}$.
It can be shown $\exists K > 0$ s.t. $|\cot(\pi z)| \le K \forall z \in \Gamma_N$, with K independent of N. The trig identity $\csc^2(\pi z) = 1 + \cot^2(\pi z)$ then implies $\left| \csc^2(\pi z) \right| \le 1 + K^2 \forall z \in \Gamma_N$. Thus for $z \in \Gamma_N$, $|z| \ge N + \frac{1}{2}$, so $|f(z)| \le \left| \frac{\cot(n\pi)}{z^2} \right| \le \frac{K}{(N + 1/2)^2} \le \frac{K}{N^2}$. Then applying the M-L Lemma gives: $\left| \int_{\Gamma_N} f(z)\ dz \right| \le \ell(\Gamma_N) \frac{K}{N^2} = \frac{4(2N+1)K}{N^2} \to 0$ as $N \to \infty$.
However, applying the CRT tells us $\int_{\Gamma_N} f(z)\ dz = 2\pi i \sum_{n=-N}^{N} \mathrm{Res}((f, n) = 2\pi i \left(2\sum_{n=1}^{N} (-1)^n \frac{1}{pin^2} + \frac{\pi}{6} \right)$ and taking $N \to \infty$ gives $\int_{\Gamma_N} = 4i \left(\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^2} + \frac{\pi^2}{12} \right)$, so $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$

Ex.: If Hol. On $\mathbb{C}$, Appropriately Bounded; Is A Polynom

Suppose h is hol. on $\mathbb{C}$ and $\exists C > 0, N \ge 0$ s.t. $|h(z)| \le C|z|^N$ for all $z \in \mathbb{C}$. Show h is a polynom of degree at most N.
The TSE for h centred at 0 is valid on $\mathbb{C}$ as h is hol. on $\mathbb{C}$. So for all $z \in \mathbb{C}$, $h(z) = \sum_{j=0}^{\infty} \frac{h^{(j)}(0)}{j1} z^j$. Let $n \ge N + 1$. We claim $h^{(n)}(0) = 0$.
By the GCIF, for $R > 0$, $\left| h^{(n)}(0) \right| = \left| \frac{n!}{2\pi i} \int_{C_R(0)} \frac{h(z)}{z^{n+1}}\ dz \right| \le \frac{n!}{2\pi} \max_{z \in C_R(0)} \left| \frac{h(z)}{z^{n+1}} \right| \ell(C_R(0)) \le \frac{n!}{2\pi} \max_{z \in C_R(0)} \left| \frac{C|z|^N}{R^{n+1}} \right| 2\pi R = \frac{n! C R^N}{R^n} = n! C R^{N-n} \to 0$ as $R \to \infty$ since $n \ge N + 1$.
Thus $h^{(n)}(0) = 0$. Then we can substitute into the above TSE to see $h(z) = \sum_{j=0}^{N} \frac{h^{(j)}(0)}{j!} z^j$, that is, that h is a polynom of degree at most N.

Ex.: Evaluating An Improper Integral Via The CRT

By CRT, calculate $I = \int_{-\infty}^{\infty} \frac{\cos(x)}{(x^2 + 4)^2}\ dx = \lim_{R\to\infty} \int_{-R}^{R} \frac{\cos(x)}{(x^2 + 4)^2}\ dx$.
I is the real part of $I_0 = \int_{-\infty}^{\infty} \frac{\exp(ix)}{(x^2 + 4)^2}\ dx$.
Let $f(z) = \frac{\exp(iz)}{(z^2 + 4)^2}$ and consider the loop Γ_R consisting of γ_R, the line along the real axis from $-R$ to R, and C_R^+, the upper semi-circle from R to $-R$. f is hol. except at the zeroes of $(z^2 + 4)^2$, $z = \pm 2i$, both of which are double poles. For $R > 2$, only $2i$ lies inside Γ_R.
$\mathrm{Res}(f, 2i) = \lim_{z \to 2i} \frac{d}{dz} ((z - 2i)^2 f(z)) = \lim_{z \to 2i} \frac{d}{dz} \left(\frac{(z - 2i)^2 \exp(iz)}{(z^2 + 4)^2} \right) = \lim_{z \to 2i} \frac{d}{dz} \left((z - 2i)^2 \frac{\exp(iz)}{(z + 2i)^2} \right) = \lim_{z \to 2i} \frac{i\exp(iz)(z + 2i)^2 - 2\exp(iz)(z + 2i)}{(z + 2i)^4} = \frac{i\exp(-2)(4i)^2 - 2\exp(-2)(4i)}{4^4} = \frac{-4ie^{-2} - 2ie^{-2}}{4^3} = \frac{-3ie^{-2}}{32}$
thus by the CRT, $\int_{\Gamma_R} f(z)\ dz = 2\pi i \mathrm{Res}(f, 2i) = 2\pi i \frac{-3ie^{-2}}{32} = \frac{3\pi e^{-2}}{16}$.
By the Jordan Lemma, $\left| \int_{C_R^+} f(z)\ dz \right| \to 0$ as $R \to \infty$.
Therefore $\int_{-\infty}^{\infty} \frac{\exp(ix)}{(x^2 + 4)^2}\ dx = \lim_{R\to\infty} \int_{-R}^{R} \frac{\exp(ix)}{(x^2 + 4)^2}\ dx = \lim_{R\to\infty} \int_{\gamma_R} f(z)\ dz = \lim_{R\to\infty} \left(\int_{\Gamma_R} f(z)\ dz - \int_{C_R^+} f(z)\ dz \right) = \lim_{R\to\infty} \int_{\Gamma_R} f(z)\ dz - \lim_{R\to\infty} \int_{C_R^+} f(z)\ dz = \lim_{R\to\infty} \frac{3\pi e^{-2}}{16} = \frac{3\pi e^{-2}}{16}$.
Hence, $\int_{-\infty}^{\infty} \frac{\cos(x)}{(x^2 + 4)^2}\ dx = I = \Re(I_0) = \Re\left(\frac{3\pi e^{-2}}{16} \right) = \frac{3\pi e^{-2}}{16}$.